

Assignment (Unit 3 & 4)

Note: last date of submission: 30/04/2026

1. Explain the basic difference between a microprocessor and a microcontroller with neat block diagrams and examples.
2. Describe embedded memory devices and external memory devices. Compare both with suitable examples.
3. Explain the features, advantages and disadvantages of CISC processors. Compare them with RISC processors.
4. Differentiate between Harvard Architecture and Von Neumann Architecture with neat diagrams.
5. Explain the architecture of the 8051 microcontroller with a labeled block diagram.
6. Describe the internal registers of 8051 and explain the function of each register.
7. Explain the memory organization of 8051 including RAM, ROM, SFR and external memory.
8. Discuss all addressing modes of 8051 with suitable examples.
9. Explain the I/O ports of 8051. Compare Port 0, Port 1, Port 2 and Port 3 in detail.
10. Describe the interrupt structure of 8051. Explain interrupt priority, enabling and disabling of interrupts.
11. Explain the working of Timer/Counter in 8051 and discuss all timer modes in detail.
12. Describe serial communication in 8051.
13. Write an assembly program to generate a unipolar square waveform of 1Khz frequency using the timer 0 of 8051 in mode 1. Assume the crystal oscillator Frequency 12 Mhz.
14. Explain communication basics in embedded systems. Discuss simplex, half duplex and full duplex modes.
15. Explain Direct Memory Access (DMA) and describe the operation of DMA controller with block diagram.
16. What is bus arbitration? Explain centralized and distributed arbitration methods.
17. Explain multilevel bus architecture with suitable diagram and advantages.
18. Explain the working of UART, SPI, and I2C protocols in detail.
19. Describe wireless communication protocols such as Bluetooth, Wi-Fi, ZigBee and RFID.
20. Explain the interfacing of LCD with 8051 microcontroller and write a program to display a message.
21. Explain stepper motor interfacing with 8051 and write a program for clockwise and anticlockwise rotation.
22. Explain ADC interfacing with microcontroller and discuss its applications in embedded systems.
23. Explain LED interfacing with 8051 and write a program for blinking LEDs.
24. Explain push button interfacing with switch debouncing technique.
25. Explain matrix keyboard interfacing with scanning method and suitable program.
26. What is latch interconnection? Explain its need and interfacing in microcontroller systems.
27. Explain Programmable Peripheral Interface (8255 PPI) with block diagram and modes of operation.